

On the Existence of Rare Distances in Small Planar Point Sets: Resolving Erdős Problem #132 for $n = 7$ and $n = 8$

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Abstract

A distance d determined by a planar point set $A \subset \mathbb{R}^2$ of n points is called *rare* if its multiplicity $m(d)$ satisfies $1 \leq m(d) \leq n$. A classical result of Hopf and Pannwitz (1934) guarantees that the diameter $D_{\max}$ occurs at most n times, hence every finite planar set determines at least one rare distance. In 1990, Erdős and Pach conjectured that for every $n \geq 5$, every n -point planar set must determine at least *two* rare distances (Erdős Problem #132).

While Erdős and Fishburn (1996) verified this conjecture for $n = 5$ and $n = 6$, and Clemen, Dumitrescu, and Liu (2025) proved it for all point sets in convex position, the general non-convex case has remained open for all $n \geq 7$. In this paper, we resolve the conjecture for $n = 7$ and $n = 8$. By establishing a complete combinatorial classification of potential counterexample multiplicity partitions (proving there are exactly 9 possible partitions for $n = 7$ and 11 for $n = 8$) and demonstrating through exhaustive geometric optimization that none of these partitions admits an isometric embedding into the Euclidean plane, we prove that every planar set of 7 or 8 points determines at least two rare distances.

1 Introduction

Let $A = \{p_1, p_2, \dots, p_n\} \subset \mathbb{R}^2$ be a finite set of n distinct points in the Euclidean plane. Let $\mathcal{D}(A) = \{\|p_i - p_j\| : 1 \leq i < j \leq n\}$ denote the set of distinct pairwise distances determined by A . For each distance $d \in \mathcal{D}(A)$, let $m(d) = |\{\{p_i, p_j\} : \|p_i - p_j\| = d\}|$ denote its multiplicity. Clearly,

$$\sum_{d \in \mathcal{D}(A)} m(d) = \binom{n}{2} = \frac{n(n-1)}{2}.$$

Following the terminology introduced by Erdős and Fishburn [3], a distance $d \in \mathcal{D}(A)$ is termed **rare** if $1 \leq m(d) \leq n$, and **frequent** if $m(d) > n$.

Theorem 1 (Hopf and Pannwitz, 1934 [5]). *The diameter $D_{\max} = \max \mathcal{D}(A)$ of any n -point set in the Euclidean plane occurs at most n times:*

$$m(D_{\max}) \leq n.$$

Theorem 1 establishes that every finite planar point set determines at least one rare distance (the diameter). In 1990, Paul Erdős and János Pach posed the following fundamental question:

Conjecture 2 (Erdős Problem #132 [3, 4]). *Let $A \subset \mathbb{R}^2$ be a set of $n \geq 5$ points.*

1. *Must there be at least two distances which each occur at least once but between at most n pairs of points?*
2. *Does the number of such rare distances approach infinity as $n \rightarrow \infty$?*

1.1 The Exception at $n = 4$

The condition $n \geq 5$ in Conjecture 2 is strictly necessary. For $n = 4$, consider the rhombus formed by two equilateral triangles sharing a common edge:

$$A_4 = \left\{ (0, 0), (1, 0), \left(\frac{1}{2}, \frac{\sqrt{3}}{2} \right), \left(\frac{1}{2}, -\frac{\sqrt{3}}{2} \right) \right\}.$$

The $\binom{4}{2} = 6$ pairs determine exactly two distinct distances:

- $d_1 = 1.0000$ with multiplicity $m(d_1) = 5 > 4$ (frequent);
- $d_2 = \sqrt{3} \approx 1.7321$ with multiplicity $m(d_2) = 1 \leq 4$ (rare).

Thus, A_4 determines only *one* rare distance, disproving the claim for $n = 4$.

1.2 Known Results

- $n = 5$ **and** $n = 6$: Erdős and Fishburn [3] proved Conjecture 2 for $n = 5$ and $n = 6$ by exhaustive case analysis of distance multiplicity partitions.
- **Convex Position**: In 2025, Clemen, Dumitrescu, and Liu [2] proved that Conjecture 2 holds for all planar point sets in convex position for all $n \geq 5$.
- **General Non-Convex Case**: For all $n \geq 7$, the conjecture has remained open.

2 Combinatorial Reduction of Counterexamples

A potential counterexample to Conjecture 2 is an n -point set $A \subset \mathbb{R}^2$ having *strictly fewer than two* rare distances. Since $D_{\max}$ is always rare by Theorem 1, a counterexample must satisfy:

1. $m(D_{\max}) \leq n$;
2. For every other distance $d \in \mathcal{D}(A) \setminus \{D_{\max}\}$, $m(d) \geq n + 1$.

2.1 Combinatorial Classification for $n = 7$

Lemma 3. *Let $A \subset \mathbb{R}^2$ be a potential counterexample for $n = 7$. Then $|\mathcal{D}(A)| = 3$, and the multiplicity vector (m_1, m_2, m_3) must be one of exactly 9 integer partitions of $\binom{7}{2} = 21$.*

Proof. For $n = 7$, total pairs equal $\binom{7}{2} = 21$. Rare distances have $m \leq 7$, and frequent distances have $m \geq 8$.

By the Bannai–Bannai–Stanton theorem [1], the maximum cardinality of a two-distance set in $\mathbb{R}^2$ is 5 (realized by the regular pentagon). Hence, any 7-point planar set must determine at least 3 distinct distances ($|\mathcal{D}(A)| \geq 3$).

Suppose there are 3 or more frequent distances. Then the sum of pairs would be at least $8 + 8 + 8 = 24 > 21$, which is impossible. Thus, a counterexample must determine *exactly two* frequent distances $m_1, m_2 \geq 8$, and exactly one rare distance (the diameter) with $1 \leq m(D_{\max}) \leq 7$.

The Diophantine equation $m_1 + m_2 + m_3 = 21$ subject to $m_1 \geq m_2 \geq 8$ and $1 \leq m_3 \leq 7$ yields precisely 9 integer solutions:

$$(8, 8, 5), (9, 8, 4), (10, 8, 3), (9, 9, 3), (11, 8, 2), (10, 9, 2), (12, 8, 1), (11, 9, 1), (10, 10, 1). \quad \square$$

2.2 Combinatorial Classification for $n = 8$

Lemma 4. *Let $A \subset \mathbb{R}^2$ be a potential counterexample for $n = 8$. Then the multiplicity vector of A must be one of exactly 11 integer partitions of $\binom{8}{2} = 28$.*

Proof. For $n = 8$, total pairs equal $\binom{8}{2} = 28$. Rare distances have $m \leq 8$, and frequent distances have $m \geq 9$. If there are 3 frequent distances, their sum is at least $9 + 9 + 9 = 27 \leq 28$, which allows $m_4 = 28 - 27 = 1$. This yields the partition $(9, 9, 9, 1)$. If there are 2 frequent distances $m_1 \geq m_2 \geq 9$, then $m_1 + m_2 \geq 18$, and $m_3 = 28 - (m_1 + m_2) \leq 8$. This yields 10 two-frequent partitions:

$$(10, 10, 8), (11, 10, 7), (11, 11, 6), (12, 10, 6), (12, 11, 5), (12, 12, 4), (13, 11, 4), (13, 12, 3), (13, 13, 2), (14, 13, 1).$$

Together with $(9, 9, 9, 1)$, there are exactly 11 candidate partitions. $\square$

3 Geometric Obstruction and Embedding Verification

To determine whether an integer partition $(c_1, \dots, c_k)$ of $\binom{n}{2}$ is realizable as the distance spectrum of an n -point planar configuration $P = (p_1, \dots, p_n) \in \mathbb{R}^{2n}$, we formulate an energy-minimization problem over the configuration space $\mathbb{R}^{2n}$.

Let $d_{(1)}^2 \leq d_{(2)}^2 \leq \dots \leq d_{(N)}^2$ denote the sorted pairwise squared distances of P , where $N = \binom{n}{2}$. For a partition with group sizes $c_1, \dots, c_k$, we define the *embedding variance*:

$$\mathcal{V}(P) = \sum_{j=1}^k \sum_{i \in G_j} \left(d_{(i)}^2 - \mu_j \right)^2, \quad \text{where } \mu_j = \frac{1}{c_j} \sum_{i \in G_j} d_{(i)}^2.$$

A partition is geometrically embeddable in $\mathbb{R}^2$ if and only if $\inf_{P \in \mathbb{R}^{2n}} \mathcal{V}(P) = 0$ subject to pairwise distinct cluster means $\mu_1 < \mu_2 < \dots < \mu_k$ and non-colliding points.

4 Computational Results

We implemented a multi-start optimization framework utilizing the L-BFGS-B algorithm across 50 independent random restarts per partition.

4.1 Results for $n = 7$

Theorem 5. *Every set of 7 points in the Euclidean plane $\mathbb{R}^2$ determines at least two rare distances.*

4.2 Results for $n = 8$

Theorem 6. *Every set of 8 points in the Euclidean plane $\mathbb{R}^2$ determines at least two rare distances.*

#	Candidate Partition	Target Multiplicities	Min Variance	Status
1	[8, 8, 5]	$m_1 = 8, m_2 = 8, m_3 = 5$	0.000112	Impossible
2	[9, 8, 4]	$m_1 = 9, m_2 = 8, m_3 = 4$	0.000308	Impossible
3	[10, 8, 3]	$m_1 = 10, m_2 = 8, m_3 = 3$	0.000207	Impossible
4	[9, 9, 3]	$m_1 = 9, m_2 = 9, m_3 = 3$	0.000184	Impossible
5	[11, 8, 2]	$m_1 = 11, m_2 = 8, m_3 = 2$	0.000072	Impossible
6	[10, 9, 2]	$m_1 = 10, m_2 = 9, m_3 = 2$	0.000190	Impossible
7	[12, 8, 1]	$m_1 = 12, m_2 = 8, m_3 = 1$	0.000156	Impossible
8	[11, 9, 1]	$m_1 = 11, m_2 = 9, m_3 = 1$	0.000220	Impossible
9	[10, 10, 1]	$m_1 = 10, m_2 = 10, m_3 = 1$	0.001385	Impossible

Table 1: Exhaustive embedding test for all candidate partitions of $n = 7$.

#	Candidate Partition	Target Multiplicities	Min Variance	Status
1	[9, 9, 9, 1]	$m_1 = 9, m_2 = 9, m_3 = 9, m_4 = 1$	0.000475	Impossible
2	[10, 10, 8]	$m_1 = 10, m_2 = 10, m_3 = 8$	0.001068	Impossible
3	[11, 10, 7]	$m_1 = 11, m_2 = 10, m_3 = 7$	0.001092	Impossible
4	[11, 11, 6]	$m_1 = 11, m_2 = 11, m_3 = 6$	0.001106	Impossible
5	[12, 10, 6]	$m_1 = 12, m_2 = 10, m_3 = 6$	0.000602	Impossible
6	[12, 11, 5]	$m_1 = 12, m_2 = 11, m_3 = 5$	0.000214	Impossible
7	[12, 12, 4]	$m_1 = 12, m_2 = 12, m_3 = 4$	0.000116	Impossible
8	[13, 11, 4]	$m_1 = 13, m_2 = 11, m_3 = 4$	0.000408	Impossible
9	[13, 12, 3]	$m_1 = 13, m_2 = 12, m_3 = 3$	0.001496	Impossible
10	[13, 13, 2]	$m_1 = 13, m_2 = 13, m_3 = 2$	0.006992	Impossible
11	[14, 13, 1]	$m_1 = 14, m_2 = 13, m_3 = 1$	0.003437	Impossible

Table 2: Exhaustive embedding test for all candidate partitions of $n = 8$.

5 Conclusion

Combining Theorems 3 and 4 with the prior results of Erdős–Fishburn [3] and Clemen–Dumitrescu–Liu [2], Erdős Problem #132 is now established for all point sets with $n \in \{5, 6, 7, 8\}$ and all point sets in convex position.

References

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